

Math 372: Homework 10

Due Monday, April 20

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1 (Central Limit Theorem). Let S_{100} be the number of heads that turn up in 100 tosses of a fair coin. Use the Central Limit Theorem to estimate

$$\mathbb{P}[S_{100} \leq 45] \quad \text{and} \quad \mathbb{P}[45 < S_{100} < 55].$$

Problem 2. A true-false examination has 48 questions. June has probability $3/4$ of answering a question correctly. April just guesses on each question. A passing score is 30 or more correct answers. Compare the probability that June passes the exam with the probability that April passes it.

Problem 3. A noodle machine in Spumoni's spaghetti factory makes about 5% defective noodles even when properly functioning. The noodles are then packed in crates containing 1900 noodles each. A crate is examined and found to contain 115 defective noodles. What is the approximate probability of finding at least this many defective noodles if the machine is properly adjusted? If you were Spumoni, would hire a technician to check out the machine?

Problem 4 (Confidence Interval). Consider the broccoli poll example discussed in lecture 26, where we considered the problem of estimating the proportion of people in Hawaii who like broccoli. In that case, we sampled $n = 100$ people and found that 20 of them like broccoli, giving the point estimate of $\hat{p} = 0.2$. We then computed a 95% confidence interval $(\hat{p} - \epsilon, \hat{p} + \epsilon)$, where $\epsilon = 0.098 = 9.8\%$ was the margin of error.

If we increase our sample size n , we expect the margin of error will decrease. How many people would we need to poll to reduce the margin of error to 3%?

Problem 5. A sample of n downed Russian *Sukhoi Su-34* fighter/bomber strike aircraft have serial numbers $x_1, x_2, \dots, x_n$. The Ukrainian Main Directorate of Intelligence knows that the aircraft were numbered consecutively at the factory starting with number α and ending with number β , so that the number of planes manufactured is

$$N = \beta - \alpha + 1.$$

For example, if $\alpha = 17$ and $\beta = 29$, then that would mean 13 planes (having serial numbers 17, 18, 19, $\dots$, 29) were manufactured. However, the Ukrainians does not know the true values of α or β . A statistician suggests using the estimator

$$\hat{N} = \max(x_1, \dots, x_n) - \min(x_1, \dots, x_n) + 1$$

to estimate the total number of planes.

- (a) If $n = 5$, $x_1 = 388$, $x_2 = 455$, $x_3 = 238$, $x_4 = 220$, and $x_5 = 431$, what is the corresponding estimate?
- (b) Will the estimate $\hat{N}$ ever be larger than the true value N ?
- (c) Do you think the estimator is unbiased for estimating N ? Explain in one or two sentences.

Problem 6. A survey collects health information on an annual basis. Here is a sample of 20 observation on HDL cholesterol level from the survey.

34	48	53	53	66	52	52
46	87	35	45	34	40	44
38	62	96	34	29	47	

The values are expressed in units of *milligrams per deciliter*. Note that HDL is “good cholesterol”; the higher its value, the lower the risk of heart attack or stroke.

- (a) Calculate a point estimate of the population mean HDL cholesterol level.

- (b) Making no assumptions about the shape of the population distribution, calculate a point estimate that separates the largest 50% of HDL levels from the smallest 50%.
- (c) Calculate a point estimate of the population standard deviation.
- (d) An HDL level of at least 60 is considered desirable as it corresponds to a significantly lower risk of heart disease. Making no assumptions about the shape of the population distribution, estimate the proportion of the population having an HDL of at least 60.

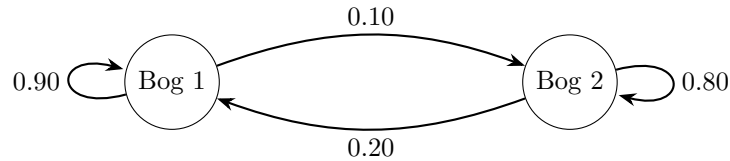
Problem 7 (Exercise 5.1.12). Two components of a computer have the following joint pdf for their useful lifetimes X and Y :

$$f(x, y) = \begin{cases} xe^{-x(1+y)} & : x \geq 0 \text{ and } y \geq 0 \\ 0 & : \text{otherwise} \end{cases}$$

- (a) What is the probability that X , the lifetime of the first component, exceeds 3?
- (b) What are the marginal pdfs of X and Y ? Are the two lifetimes independent?
- (c) What is the probability that the lifetime of at least one component exceeds 3?

Problem 8. Let X be a random variable with mean μ and variance σ^2 , and let a, b, c be arbitrary real numbers. Define a new random variable $Y := aX^2 + bX + c$. Find $\mathbb{E}[Y]$.

Problem 9 (Extra credit). There are two bogs, Bog 1 and Bog 2. Each day, a mosquito in Bog 1 has a 10% chance of migrating to Bog 2, and a 90% chance of not migrating. Similarly, each day a mosquito in bog 2 has 20% chance of migrating to bog 1, and an 80% probability of not migrating. This situation is depicted in the following diagram:



To clarify, we assume that each mosquito repeats the above migration process at the beginning of each day with no memory of prior days, and that the population of mosquitoes is constant (i.e., mosquitoes never die, new ones are never born, mosquitoes never leave the two bogs, etc).

Suppose on Day 1, 50% of the mosquitoes are in Bog 1 and 50% are in Bog 2. After a long time has passed, what proportion of the mosquito population do you expect to find in bog 1, and what proportion do you expect to find in bog 2?